

DeepLabCut-HRNet coordinate-and-descriptor route

DLC SuperAnimal pose + detector

MARS pose annotations
converted to maDLC format
21,614 source frames
43,228 animal pose examples
pose train/test 41,066/2,162
pose mAP 0.918
detector mAP50-95 0.710

Native top-down full-video cache

Detector boxes + HRNet
keypoints
assigned animal slots
group and animal crops
pose-derived relations
30,493 train/val windows

Validated HRNet feature tap

Pre-head HRNet backbone
four branches required
widths [32,64,128,256]
semantic_concat = 480-D
cache stops on
validation failure

Behavior decoding test

Same checkpoint yields
keypoints + visual descriptors
Attention-256 classifier
28 held-out videos
0 blocking failures
frame F1 0.749
bout F1@0.25 0.670

The workflow produces both coordinates and reusable visual descriptors for behavior decoding.